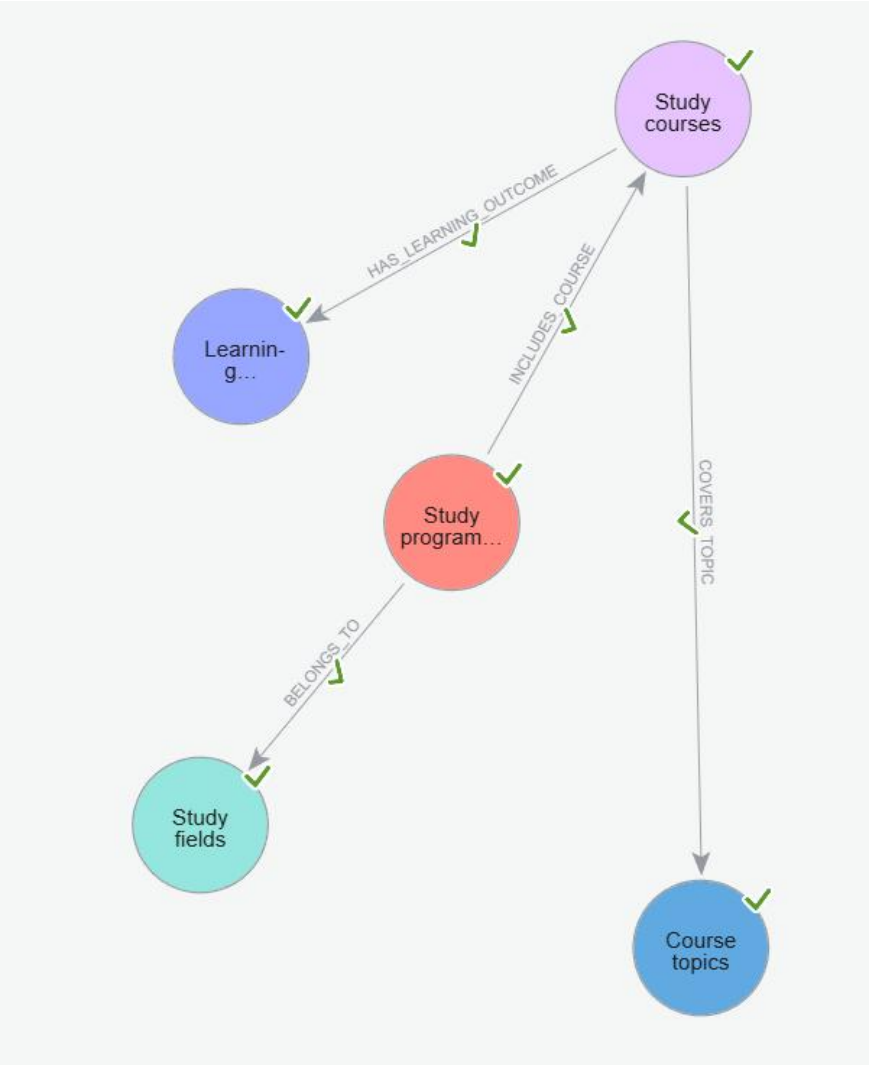
AuraDB Knowledge Graph creation

Anna Laha

071RIC181

Schema


Study courses
HAS_LEARNING_OUTCOME
Learnin-g...
INCLUDES_COURSE
Study program...
COVERS_TOPIC
BELONGS_TO
Study fields
Course topics

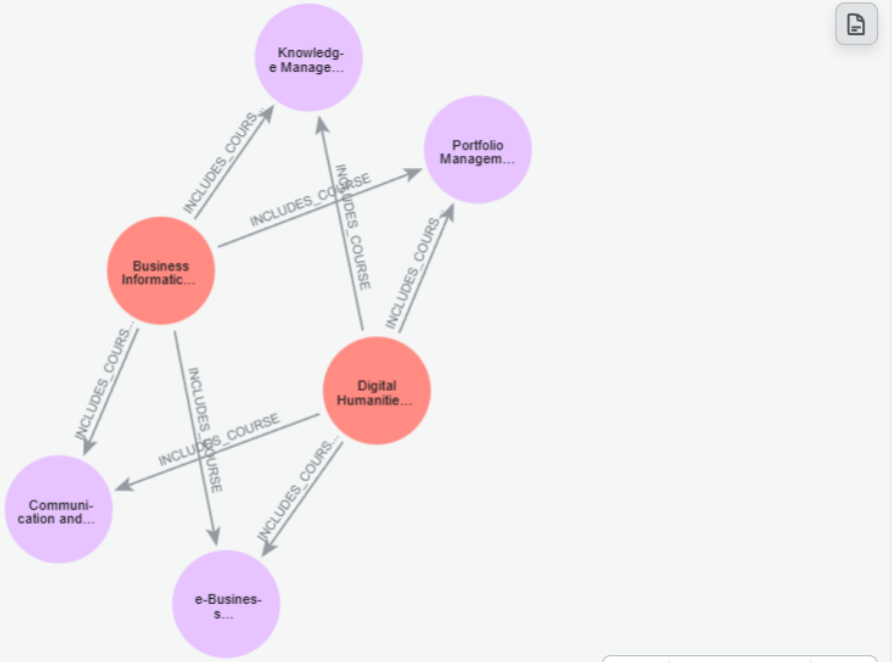
Query: Courses shared by multiple study programmes

```
1 MATCH (p)-[:INCLUDES_COURSE]->(c)
2 WITH c, count(DISTINCT p) AS programmeCount
3 WHERE programmeCount > 1
4 MATCH path = (p)-[:INCLUDES_COURSE]->(c)
5 RETURN path;
```

Run

Graph | Table | Raw

Search graph...



Results overview

Nodes (6)

* (6) | Study courses (4) | Study programms (2)

Relationships (8)

* (8) | INCLUDES_COURSE (8)

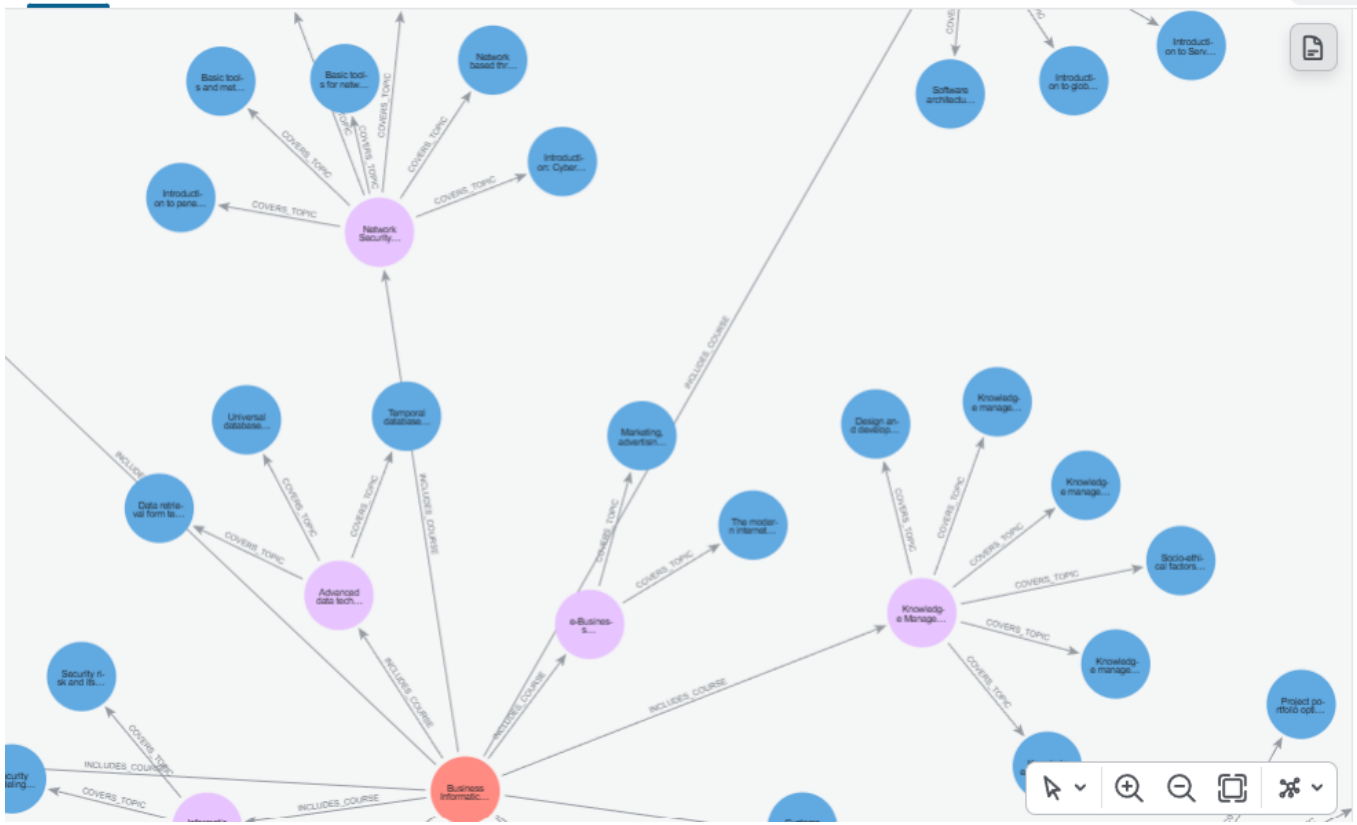
Query: Which topics are covered by the courses in each study program?

```
1 MATCH path = (programme)-[:INCLUDES_COURSE]->
2              (course)-[:COVERS_TOPIC]->(topic)
3 RETURN path
4 LIMIT 100;
```

Run

Graph Table Raw

Search graph...



Results overview

Nodes (115)

- * (115)
- Course topics (100)
- Study courses (14)
- Study programms (1)

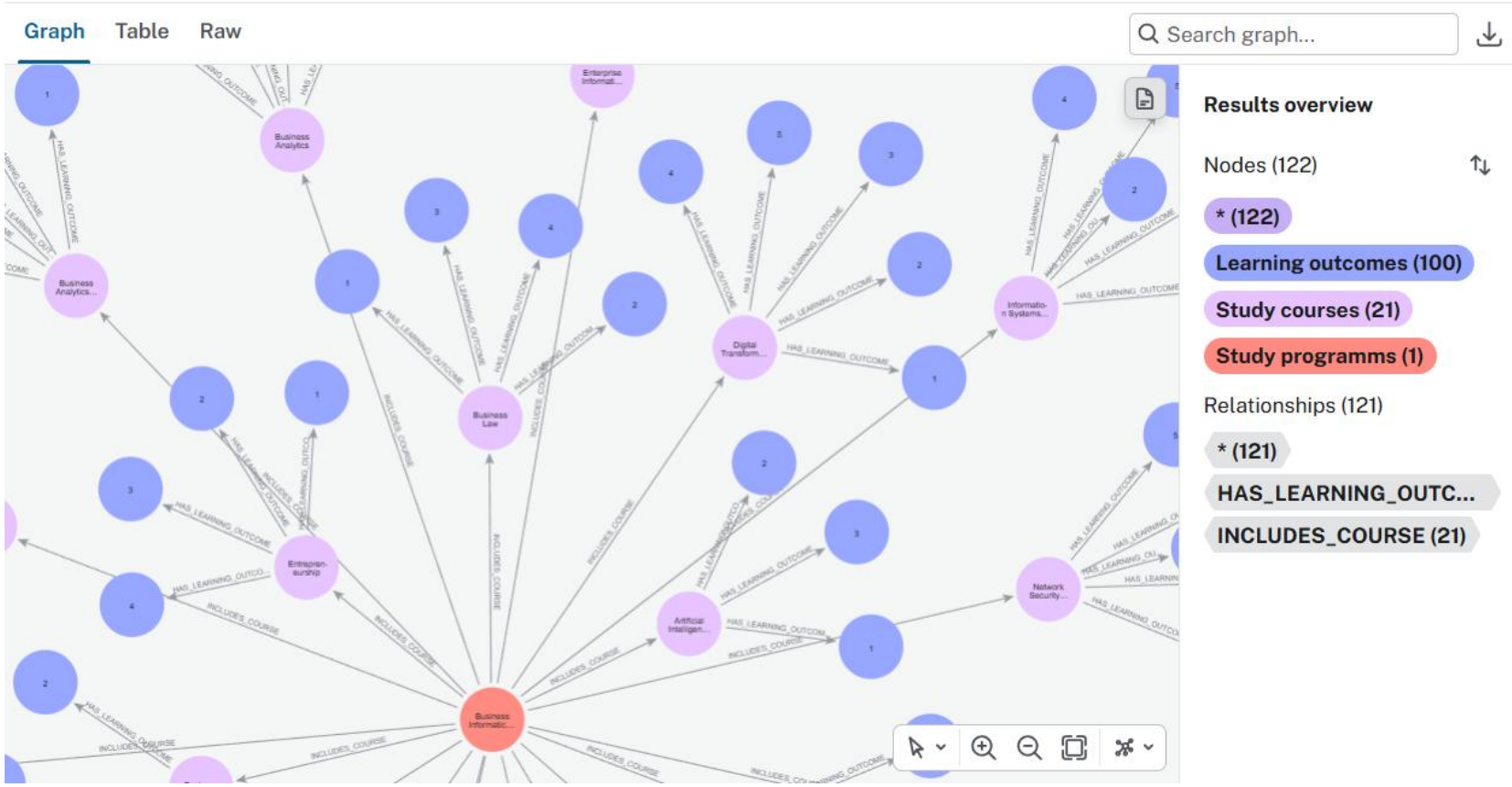
Relationships (114)

- * (114)
- COVERS_TOPIC (100)
- INCLUDES_COURSE (14)

Query: Which learning outcomes does each study program address through its courses?

```
1 MATCH path = (programme)-[:INCLUDES_COURSE]->
2              (course)-[:HAS_LEARNING_OUTCOME]->(outcome)
3 RETURN path
4 LIMIT 100;
```

Run



Query: Which study fields have the largest number of distinct courses?

```
MATCH (field)<-[:BELONGS_TO]-(programme)
             -[:INCLUDES_COURSE]->(course)
WITH field, count(DISTINCT course) AS courseCount
ORDER BY courseCount DESC
LIMIT 3
MATCH path = (field)<-[:BELONGS_TO]-(programme)
                    -[:INCLUDES_COURSE]->(course)
RETURN path;
```

Graph Table Raw

